# ScholarLens: laptop study commands

Windows PowerShell · NVIDIA RTX 3050 (6 GB) · github.com/madhureddy25/ScholarLens · same commands as LAPTOP_COMMANDS.ps1 in the repo

How to use this sheet. Run the grey commands one at a time, in number order: copy one line, paste it into PowerShell, press Enter, and wait for it to finish before the next. Each numbered line is one complete command. Copy only the grey text, not the number. If anything errors, copy the full message, open the repo in Claude Code and say continue ScholarLens. Troubleshooting: docs/WINDOWS.md.

## 0 · One-time setup

### Install the tools

```
1  winget install --id Git.Git -e
2  winget install --id Python.Python.3.12 -e
3  winget install --id GitHub.cli -e
4  winget install --id Ollama.Ollama -e
```

### Close PowerShell and open a NEW window, then check the GPU (it should list “RTX 3050”)

```
5  nvidia-smi
```

### Get the code

```
6  cd $HOME
7  git clone https://github.com/madhureddy25/ScholarLens.git
8  cd ScholarLens
```

### Create the Python environment

```
9  py -3.12 -m venv .venv
10 Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
11 .\.venv\Scripts\Activate.ps1
12 python -m pip install --upgrade pip
13 pip install torch --index-url https://download.pytorch.org/whl/cu126
14 pip install -e ".[all,dev]"
15 [Environment]::SetEnvironmentVariable("PYTHONUTF8", "1", "User")
```

### Check that it works: the first should print True NVIDIA GeForce RTX 3050 … and all tests should pass

```
python -c "import torch; print(torch.cuda.is_available(), torch.cuda.get_device_name(0))"
pytest -q -m "not slow"
```

### Download the AI models (about 9 GB in total)

```
ollama pull qwen2.5:7b
ollama pull phi3.5
ollama pull llama3.2:3b
```

Every time you open a new PowerShell window, run these two first:

```
cd $HOME\ScholarLens
.\.venv\Scripts\Activate.ps1
```

## 1 · Build the benchmark

### Smoke test (about 30 minutes): check the counts it prints

```
scholarlens bench build configs/bench.laptop.toml --n-human 40 --out data/bench/laptop-smoke
scholarlens lens train data/bench/laptop-smoke/bench.jsonl --out models/laptop-smoke
```

### Stop the laptop from sleeping while plugged in

```
powercfg /change standby-timeout-ac 0
```

### Full build, overnight. If it stops for any reason, run the SAME command again to resume.

```
scholarlens bench build configs/bench.laptop.toml
```

## 2 · DIPPER on Kaggle (free GPU)

### Zip the benchmark

```
Compress-Archive -Path data\bench\laptop-v1 -DestinationPath laptop-v1.zip -Force
```

**On kaggle.com**

1. Datasets → New Dataset: upload `laptop-v1.zip` and keep it private.
2. Code → New Notebook. In Settings, set Accelerator to GPU T4 and turn Internet on (needs a phone-verified account).
3. Add Input: pick your dataset.
4. Paste and run these notebook cells (about 3–4 hours). These are Kaggle cells, not PowerShell.

```
!git clone https://github.com/madhureddy25/ScholarLens.git
%cd /kaggle/working/ScholarLens
!pip install -q -e ".[ml]" bitsandbytes sentencepiece datasets
!ls /kaggle/input/*/
!mkdir -p data/bench && cp -r /kaggle/input/*/laptop-v1 data/bench/
!scholarlens bench build configs/bench.kaggle-dipper.toml
!mkdir -p /kaggle/working/dipper && cp data/bench/laptop-v1/transforms/dipper_*.jsonl /kaggle/working/dipper/
```

5. From the notebook's Output panel, download `dipper_L20_O0.jsonl` and `dipper_L60_O60.jsonl`.
6. Copy both files into `data\bench\laptop-v1\transforms\` on the laptop.

**Merge the DIPPER files into the benchmark (nothing gets regenerated)**

```
scholarlens bench build configs/bench.laptop.toml
```

## 3 · Train Lens and Quill-RL

**Lens: the main run, then the two comparison runs for the paper**

```
scholarlens lens train data/bench/laptop-v1/bench.jsonl --out models/lens-v1 --gnn
scholarlens lens train data/bench/laptop-v1/bench.jsonl --out models/lens-unseen-dipper --exclude ai+dipper_L60_O60
scholarlens lens train data/bench/laptop-v1/bench.jsonl --out models/lens-no-edits --exclude "human+quill_proofread,human+quill_clarity"
```

**Quill-RL: first free the GPU memory Ollama is holding**

```
ollama stop qwen2.5:7b
```

**Then train (1–3 hours). It prints whether the new version was ACCEPTED: note the answer**

```
scholarlens learn quill --bench data/bench/laptop-v1/bench.jsonl --base Qwen/Qwen2.5-0.5B-Instruct --mode proofread --steps 200 --out models/quill-rl-v1
```

**Try the trained editor on the sample paper**

```
scholarlens edit tests/fixtures/sample_paper.tex -m quill:models/quill-rl-v1 --mode proofread
```

## 4 · Publish the first releases

### Log in to GitHub and set up git (put your own name and email in the first two)

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
gh auth login
gh auth setup-git
```

### Publish

```
scholarlens learn publish bench data/bench/laptop-v1
scholarlens learn publish lens models/lens-v1 --force
```

### Publish Quill ONLY if step 3 printed “accepted”

```
scholarlens learn publish quill models/quill-rl-v1
```

### Start the first automatic Lens retraining on GitHub

```
gh workflow run learn --repo madhureddy25/ScholarLens
```

## 5 · Save the results for the paper

### Copy the reports into the repo and push them

```
New-Item -ItemType Directory -Force reports
Get-ChildItem models -Directory | ForEach-Object { if (Test-Path "$($_.FullName)\report.md") { Copy-Item "$($_.FullName)\report.md" "reports\$($_.Name).md" } }
Copy-Item data\bench\laptop-v1\manifest.json reports\bench-manifest.json
Copy-Item models\quill-rl-v1\learn_report.json reports\quill-rl-v1.json
git add reports
git commit -m "Laptop study results"
git push
```

Then: open the repo in Claude Code and say continue ScholarLens: draft the paper from reports/.

### Turn sleep back on when everything is finished

```
powercfg /change standby-timeout-ac 30
```

## If something fails

| | |
|---|---|
| Activate.ps1 cannot be loaded | Run command 10 (Set-ExecutionPolicy) and answer Y. |

| | |
|---|---|
| torch.cuda.is_available() is False | Update the NVIDIA driver (GeForce Experience or nvidia.com), then run command 13 again. |
| Ollama connection refused | Start the Ollama app from the Start menu, or run `ollama serve` in another window. |
| CUDA out of memory in learn quill | Run `ollama stop qwen2.5:7b` first; also try adding `--group 2`. |
| A build stopped halfway | Run the same command again: every step is cached and resumes. |
| Anything else | Copy the full error message and ask Claude in Claude Code with the repo open. |